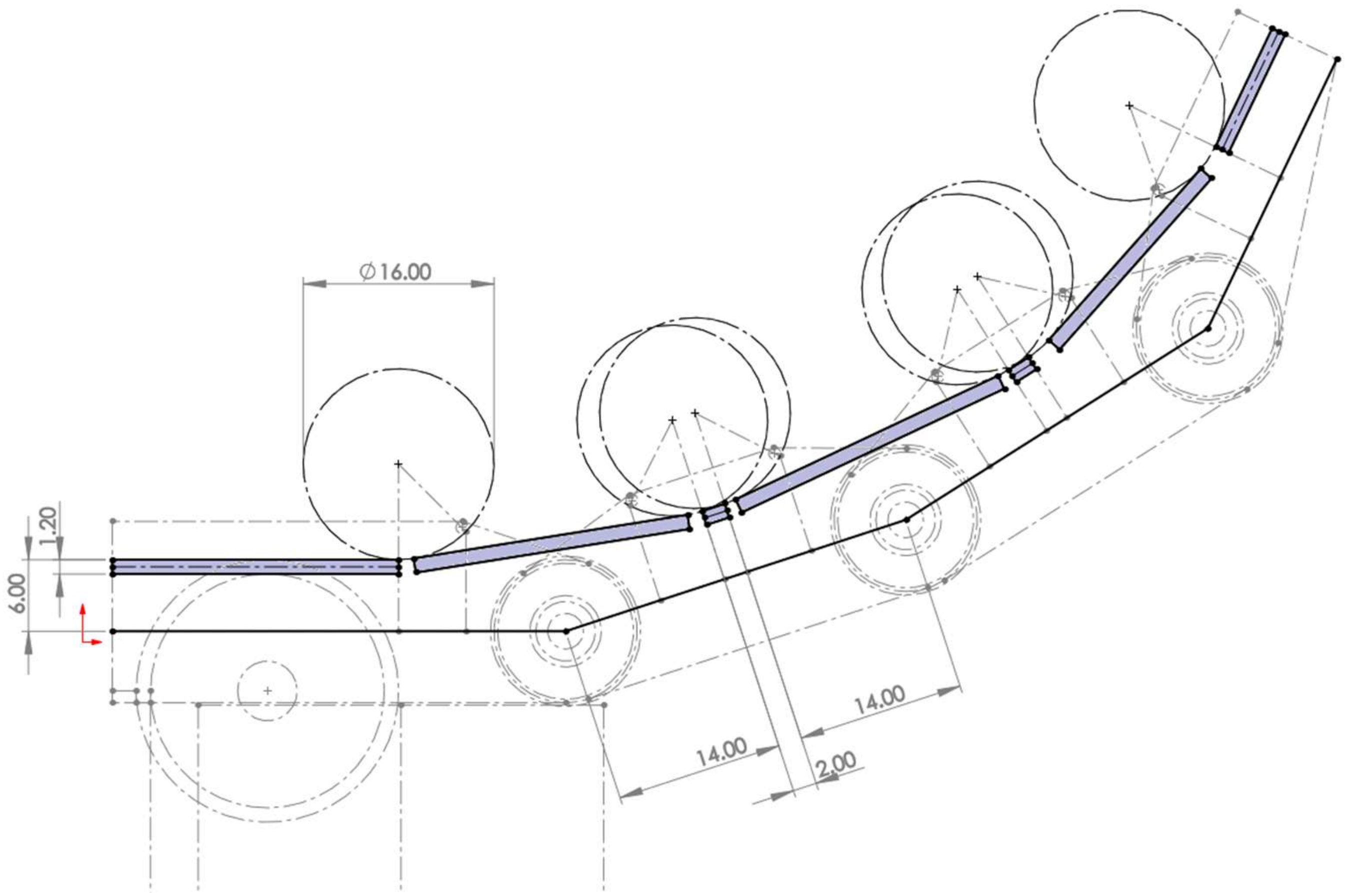


1. Starting with the tendons and link lengths. Tendon diameter is 0.8mm. Strain limit for nitinol wire at 5%.

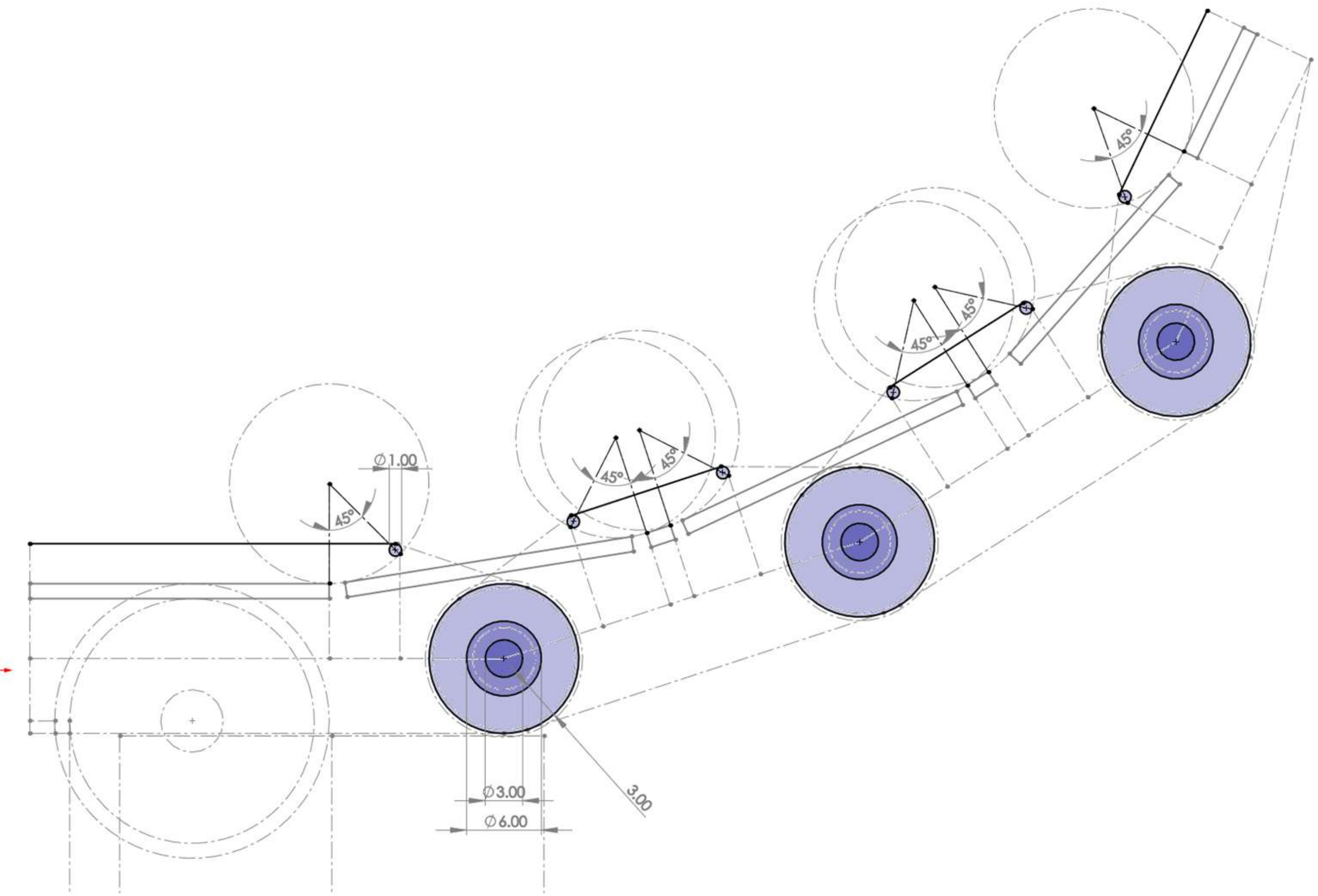
   Dm=d((1/e)-1)

   Dm= bend diameter
   d= wire diameter
   e=strain

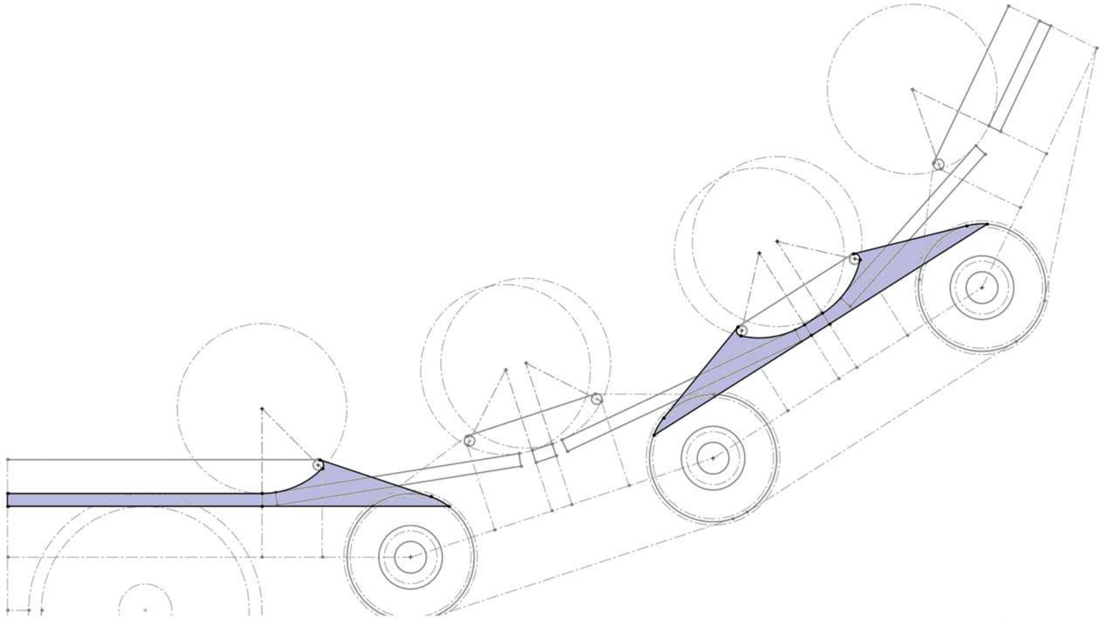
   Dm=0.8((1/0.05)-1)
   Dm=15.2mm

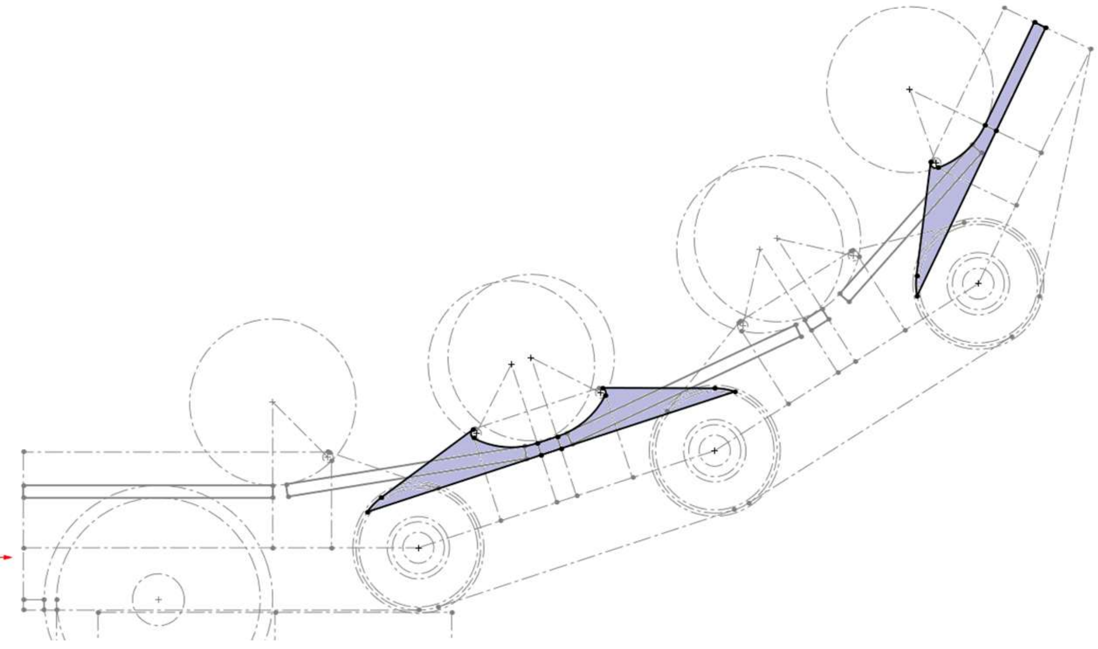
   Rounded up to 16mm bend diameter.

2. Tendon is 0.8mm diameter, added 0.2mm either side for free sliding tolerance with 3D printed parts
3. Link lengths calculated by:
   = 6 + (16/2) + 2 + 6 + (16/2)
   = 14 + 2 + 14
   = 30mm
   The 2mm can be adjusted to extend link lengths as required. The 14mm dimension keeps the tendon channels aligned from link to link.

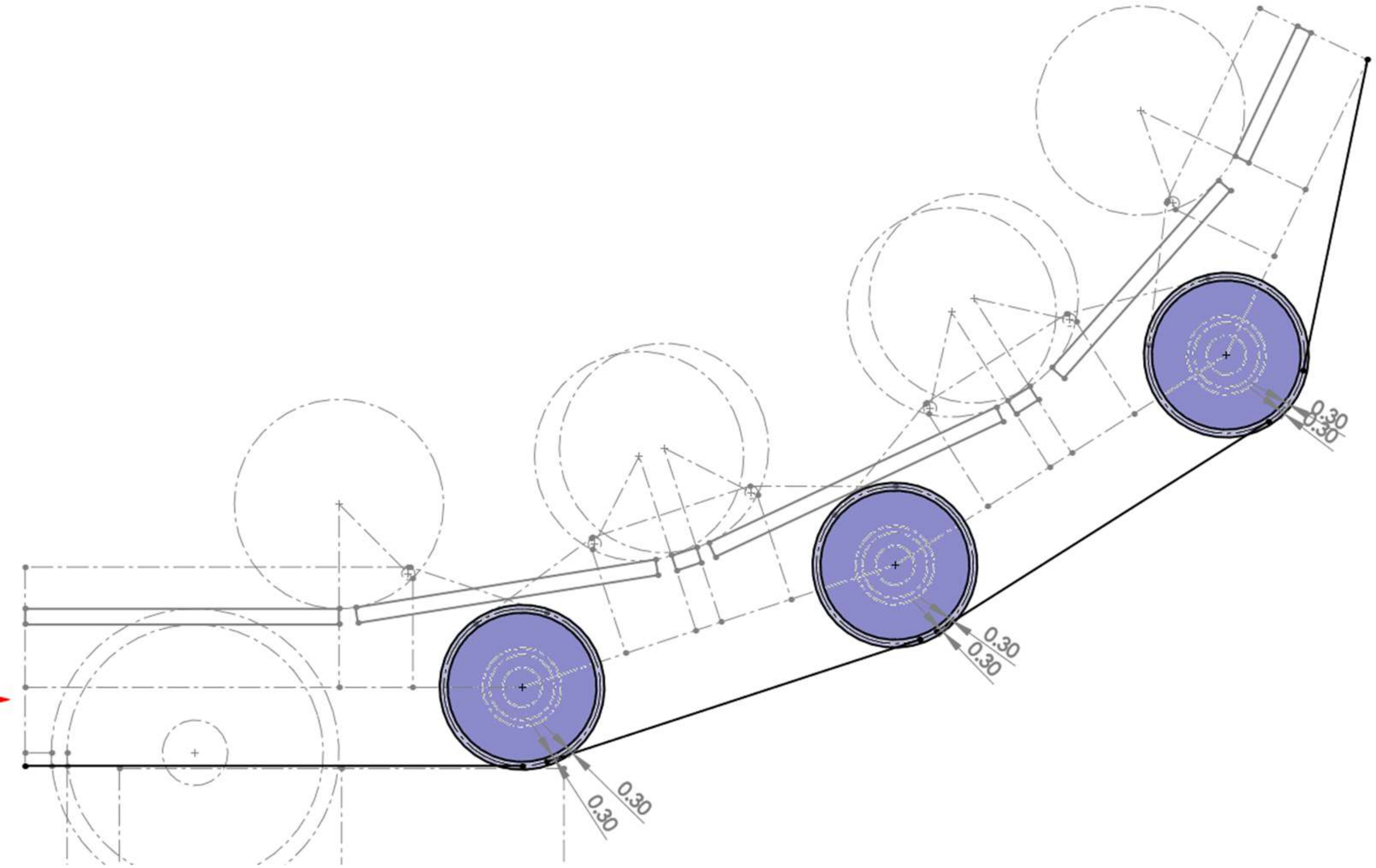


1. Adding in bearings and dimensioning the tendon routing.
2. Bearings are 3mm inner, 6mm outer, 2.5mm wide.
3. 45 degree in place as the links are limited to 90 degrees.
4. 1mm fillet added to remove sharp edges on the tendons.
5. Inner surface of finger added.

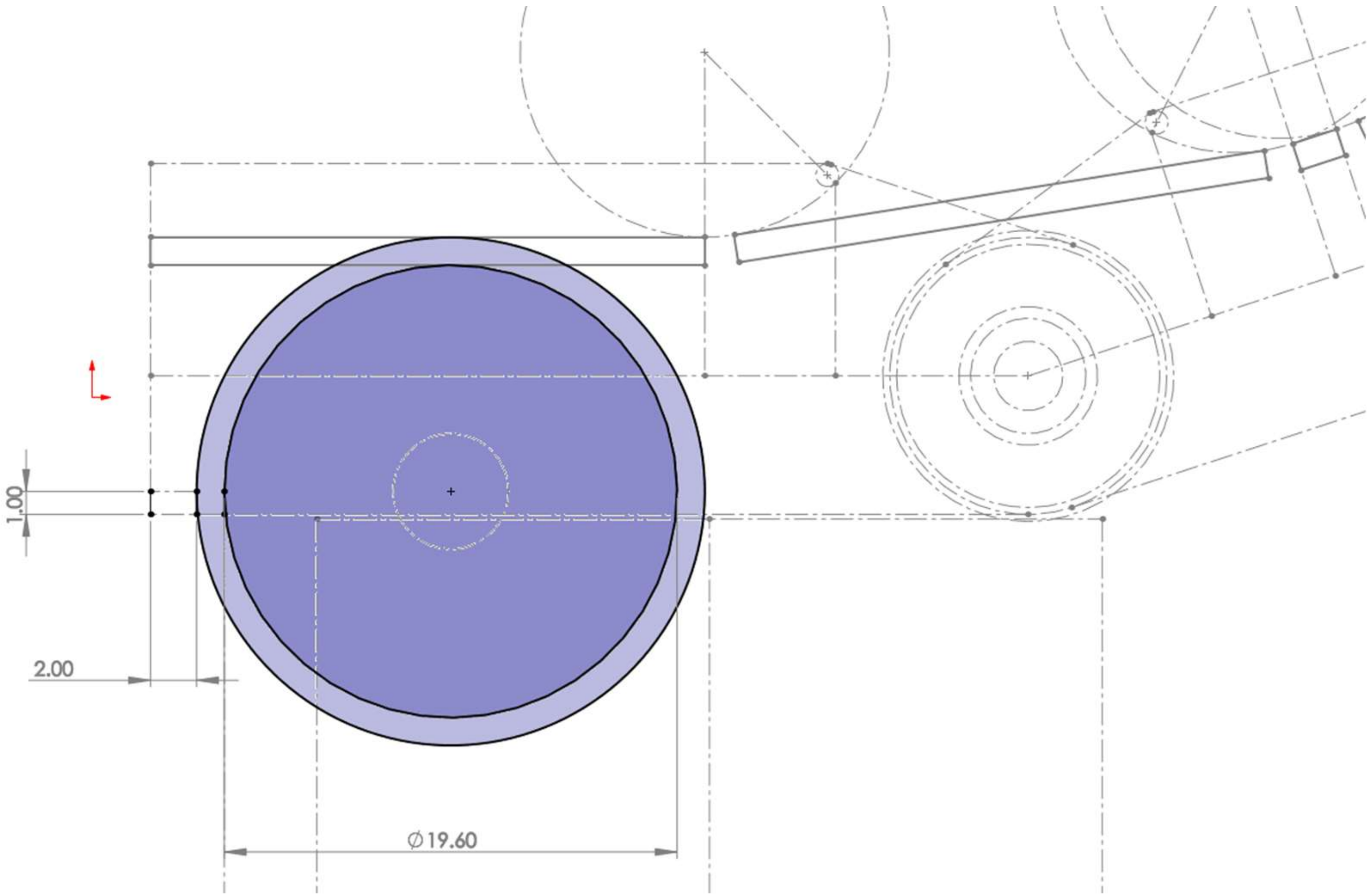




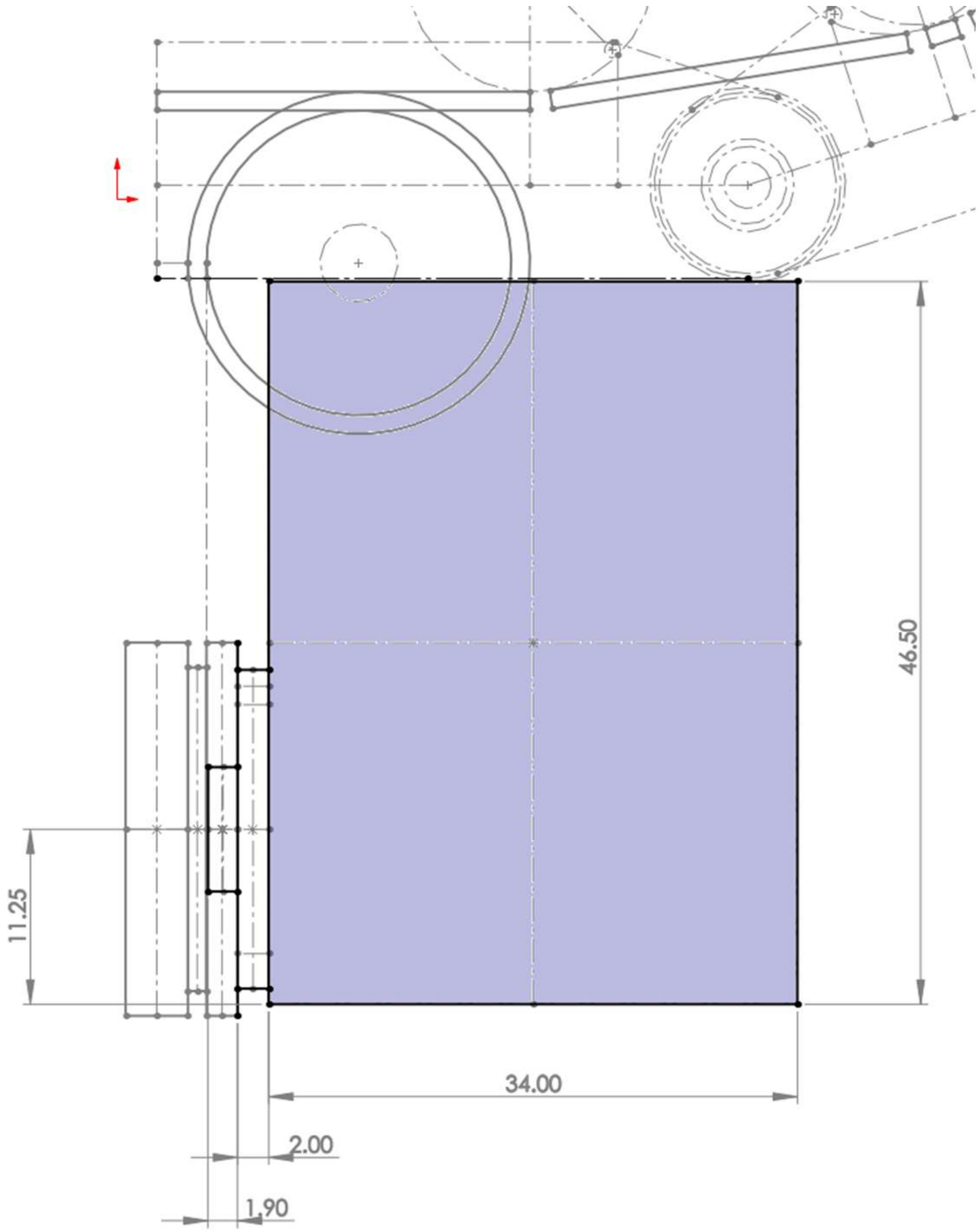
1. Tendon channel within each link completed



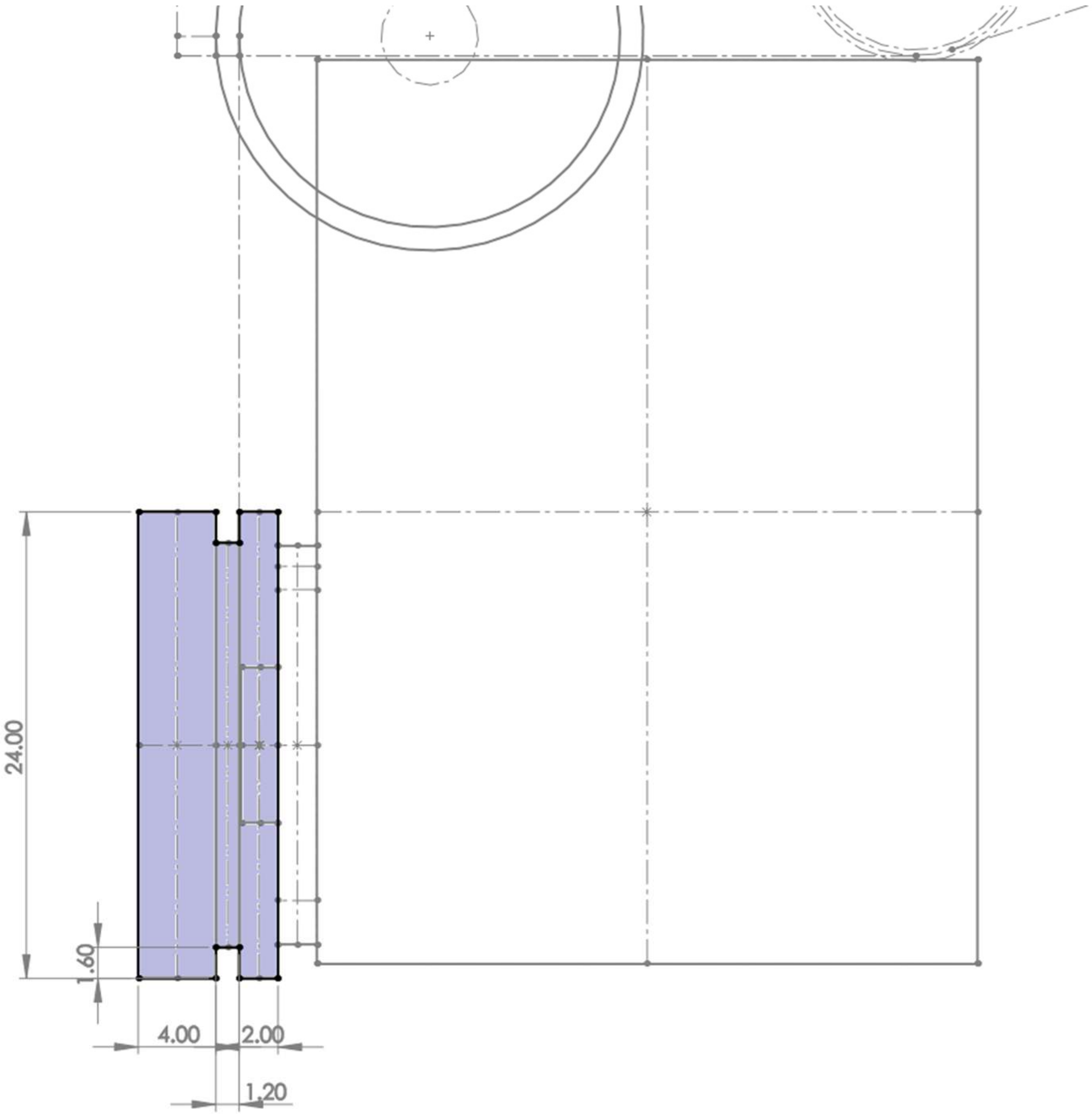
1. 0.3mm clearance added to where links would have touched.
2. Outer surface of finger added.



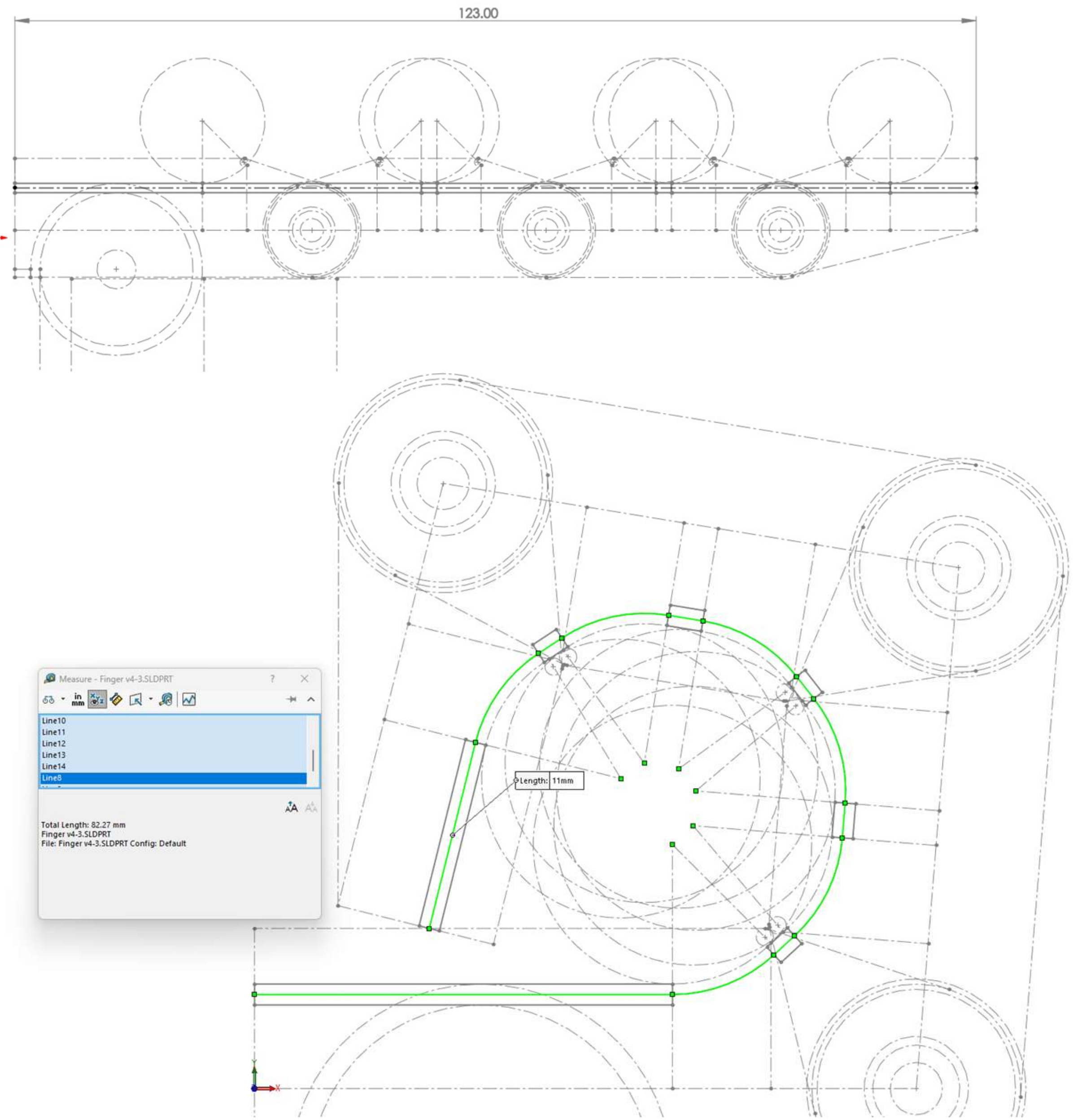
1. Added tendon channel to redirect it below finger.
2. 2mm used for minimum wall thickness
3. 1mm used to straighten the channel before exiting.
4. 19.60mm bending diameter on tendon does not exceed the 5% strain limit.



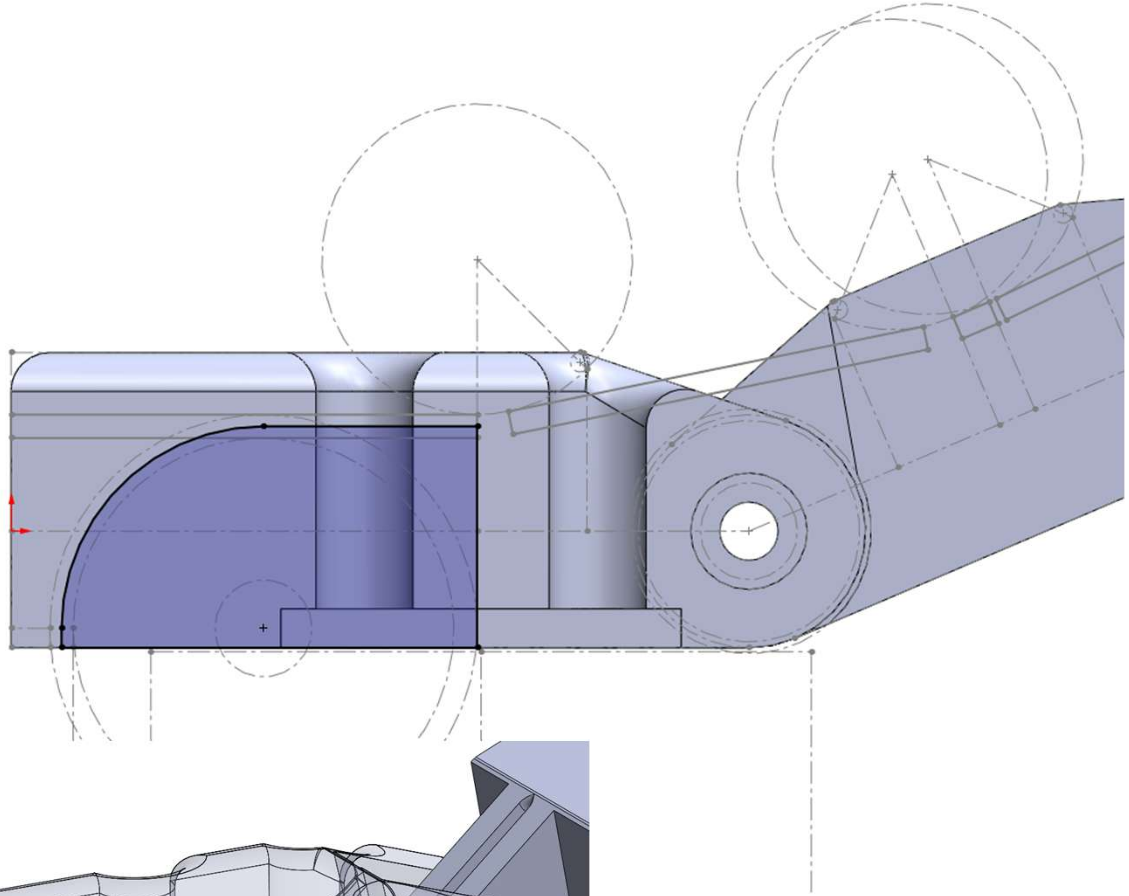
1. Dynamixel XM430-W250
   https://emanual.robotis.com/docs/en/dxl/x/xl430-w250/

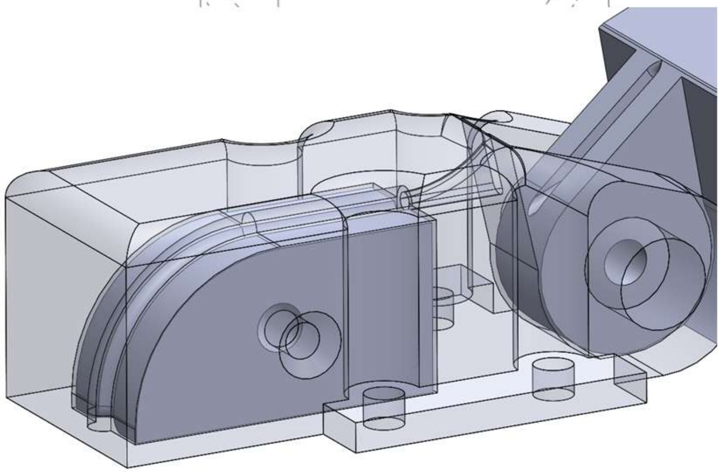


1. Designing pulley to align with tendon output
2. 2mm min wall thickness
3. 4mm due to length of brass threaded inserts.
4. 1.2mm to match tendon channel width
5. 1.6mm due to the mounting holes to connect pulley to motor horn.
6. 24mm to due aim to keep pulley diameter smaller than motor width.

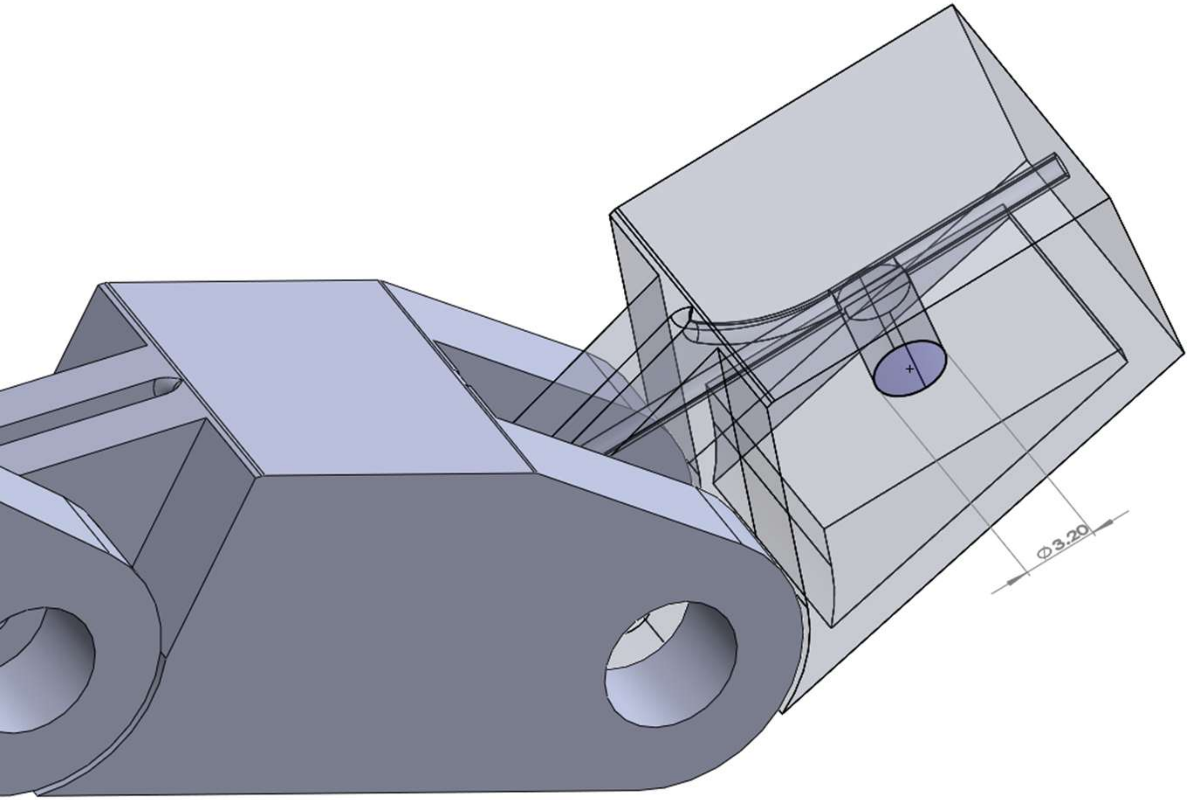


1. Tendon length at max extension: 123mm
2. Tendon length at min extension: 82.27mm
3. Change in length: 40.73mm
4. Pulley diameter: 20.8mm
   Pulley circumference: 20.8 x 3.14 = 65.31mm
5. Change in tendon length within range of pulley.
6. If the pulley circumference isn't large enough, the pulley will extend in length to wind the tendon in a spiral pattern, so the pulley diameter doesn't have to increase.

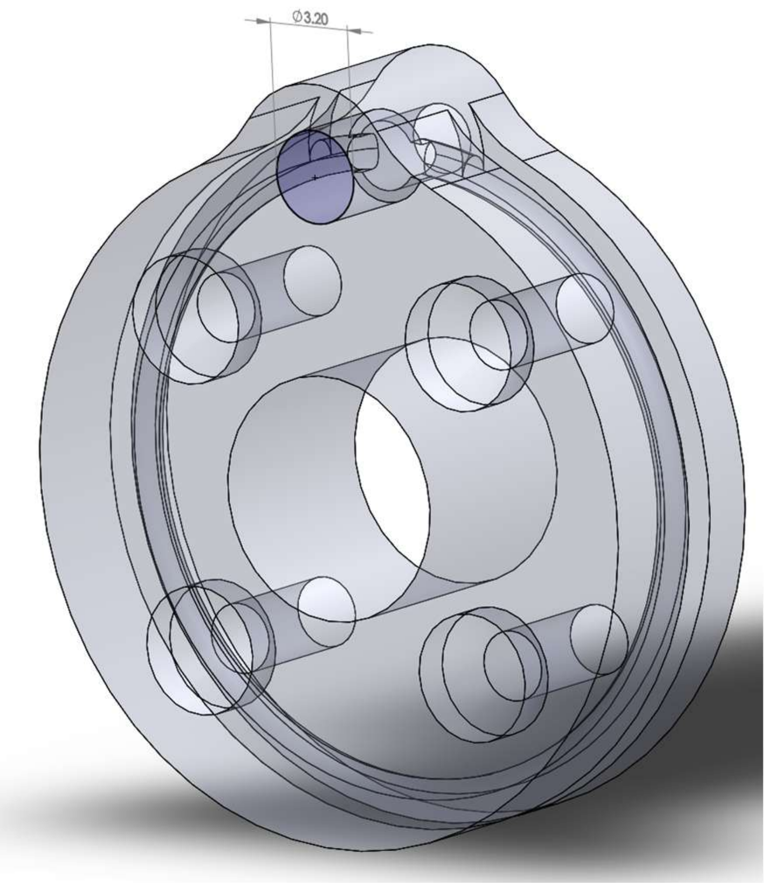




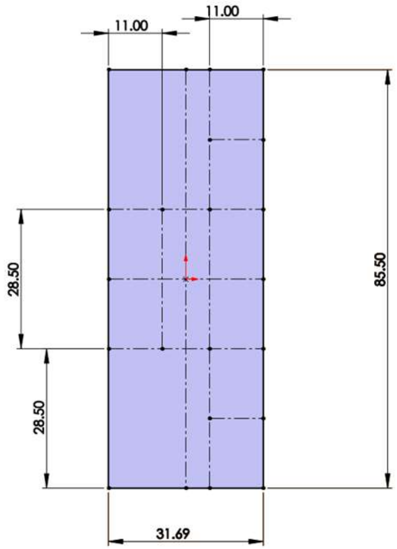
1. Assembly is complicated when installing the tendon which had to be pushed through the 90 degree bend. The sharp edge of the wire dug into the plastic rather than follow the channel.
2. A removable plug was built in to allow the tendon to be installed then the plug completes the tendon channel.

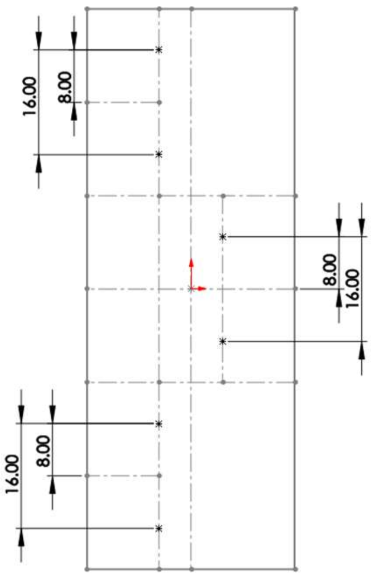


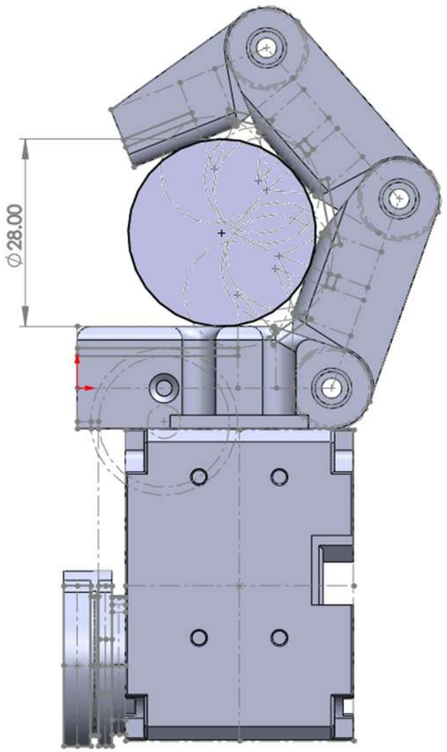
1. Tendon held in place at the distal link with a M2 bolt.
2. M2 Bolt held in place with a brass threaded insert that is pressed into the 3.20mm hole.



1. Tendon held in place at the pulley with a M2 bolt.
2. M2 Bolt held in place with a brass threaded insert that is pressed into the 3.20mm hole.







1. To mount multiple fingers, we screw the 3 motors onto a single baseplate.
2. The position of the motor on the base plate was driven by the aligning the item we're holding with each finger.